English Medium

EDU TERIA

72nd BPSC

Prelims Online Batch

BIOLOGY

Topic wise Notes

LECTURE 29

Genetics and Evolution

Genetics and Evolution

Genetics

Genetics is the branch of biology that studies heredity and the Variation of inherited characteristics.

Fundamentals of Genetics

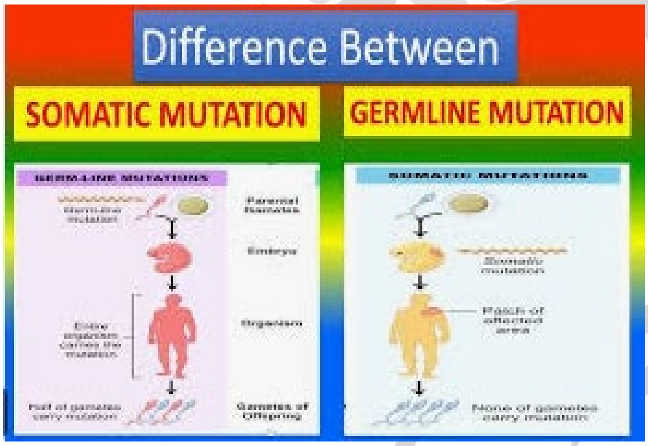
- Heredity refers to the transmission of traits from one generation to the next.
- Term given by William Bateson
- Father of Genetics: Gregor Mendel
- Unit of heredity → Gene
- Location of genes → Chromosomes (in nucleus)

Variation

Variation refers to the differences among individuals of a species.

Types of variation:

1. Genetic variation
2. Environmental variation



Chromosome

Chromosomes are thread-like structures present in the nucleus, made of DNA and proteins (histones), which carry genes (hereditary information) from one generation to another.

Chromosome (23 pairs)		
Autosome	Sex Chromosome	
(22 pairs)	Male (XY) (Responsible for sex determination)	Female (XX)
Somatic chromosome	One pair (Heterozygous)	Homozygous

Chromosomal Aberration/Abnormality

1. **Down Syndrome**—Trisomy of 21st pair of chromosomes (Total chromosome = 47)
   Eg. Mental retardation, Flat palm, abnormal neck.
2. **Adward Syndrome**— Trisomy of 18th pair of (47 chromosome) chromosome.
3. **Patou Syndrome**— Trisomy of 13th pair of chromosomes (47 chromosomes)
4. **Klinefelter Syndrome**— (22+XXY)– (extra sex chromosome). Male Character with feminine sterile in nature. Female sex character also present.
5. **Turner Syndrome**— (22 pair + XO) (Total chromosome = 45). Female sterile, rudimentary ovary.

Autosomal recessive gene mutation Disorder

Common Disorders:

- **Cystic Fibrosis (CF):** Affects mucus-producing glands.
- **Sickle Cell Disease (SCD):** Impacts hemoglobin production.
- **Tay-Sachs Disease:** Metabolic disorder damaging nerve cells.
- **Phenylketonuria (PKU):** Affects amino acid metabolism.
- **Thalassemia:** Blood disorder causing low hemoglobin.
- **Albinism:** Affects melanin production.

X-linked recessive disease:

Gene defect present on X chromosome.

1. **Hemophilia**

   Blood clotting absent because of defecting gene responsible for formation blood clotting factor.

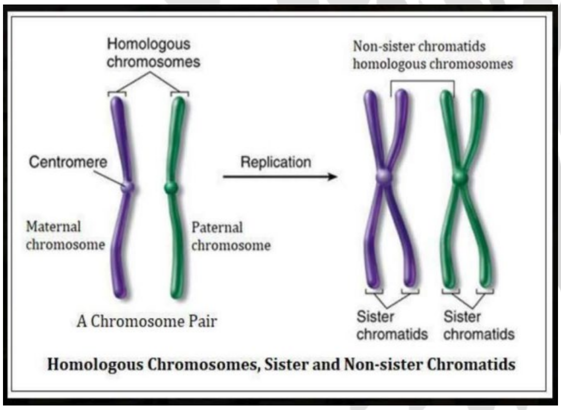
   - **Aemophilic male** — X^nY Genotype
   - **Hemophilic female** — X^nX^n (not possible) lethal in nature. Death in embryo stage.
   - **Carrier female** (not Hemophilic) — X^nX
   Vector Male

2. **Colour blindness**
'X' linked recessive disorder.
Not able to differentiate between primary colour. (Red, blue Green)
Colour blind male — X^cY
Colour blind female — X^cX^c
Carrier female — X^cX

Gregor Mendel:

- Genetics Gregor Mendel is known as the "Father of Genetics" for his foundational experiments with garden pea plants (Pisum sativum).
- The term "genetics" was coined by William Bateson in 1906

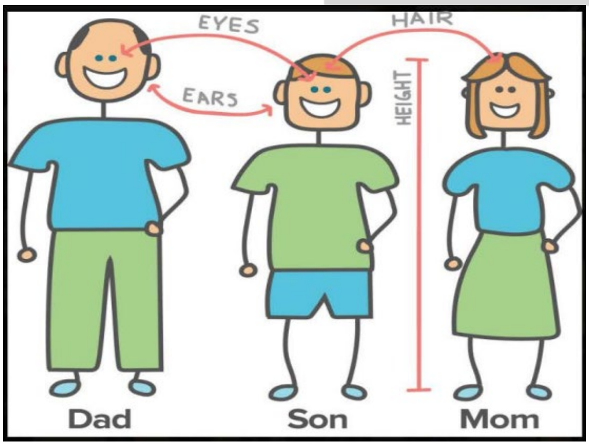
Homologous Chromosome



Homologous Chromosomes, Sister and Non-sister Chromatids

- **Gene:** The basic unit of heredity, a specific segment of DNA that provides instructions for a particular trait or protein. one gene-one enzyme
- **Allele:** An alternative form or variation of a gene (e.g., the gene for height alleles for "tall" and "dwarf").

Characteristic

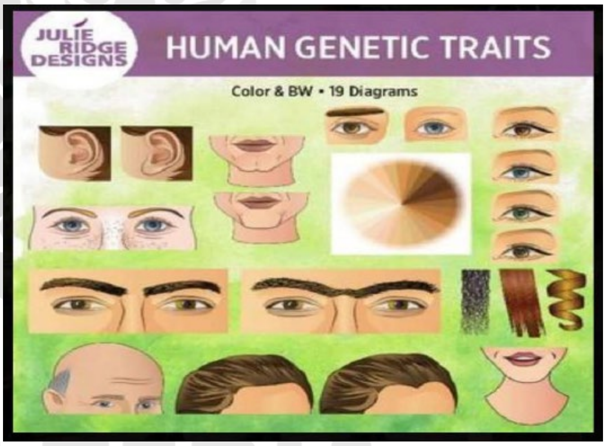


Genetic Terminology

- **Chromosome:** Thread-like structures located inside the nucleus of a cell, made of DNA coiled around proteins. itistone protein
- **Humans** have 23 pairs (46 total) of chromosomes.
- **DNA (Deoxyribonucleic acid):** The genetic material in most living organisms, possessing a double-helix structure discovered by Watson and Crick.

Trait

Character	
Traits ↓ Two allele	**Multiple traits** ↓ Multiple allele



- **Genotype:** The genetic constitution or set of genes an organism possesses for a specific trait.
- **Phenotype:** The observable, physical, or biochemical characteristics of an organism, resulting from the interaction of its genotype and the environment.
- **Dominant Trait/Allele:** An allele that expresses its effect even in the presence of Follow Us f an alternative (recessive) allele (e.g., 'Tall' in pea plants).
- **Recessive Trait/Allele:** An allele whose effect is masked in the presence of a dominant allele and is only expressed when two copies are present.
- **Homozygous:** Having two identical alleles for a particular gene (e.g., TT or tt).

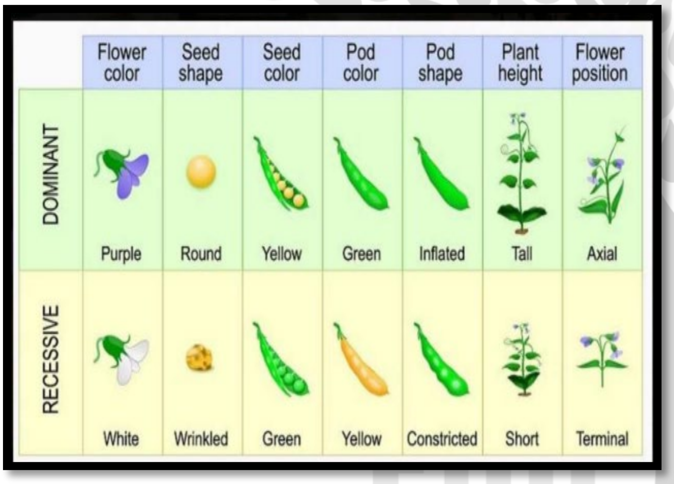
- **Heterozygous:** Having two different alleles for a particular gene (e.g., Tt). Mendel's Laws of Inheritance

Mendel's work established the fundamental laws of heredity:

- **Law of Dominance:** In a cross between two different organisms for a single trait, only one form of the trait (the dominant one) appears in the first generation (F1).
- **Law of Segregation (Purity of Gametes):** During the formation of gametes (sex cells), the two alleles for a character separate or segregate from each other so that each gamete receives only one allele.

Mendelism

Mendel studied 7 characters or 7 pairs of contrasting traits:



Mendel's Law of Inheritance

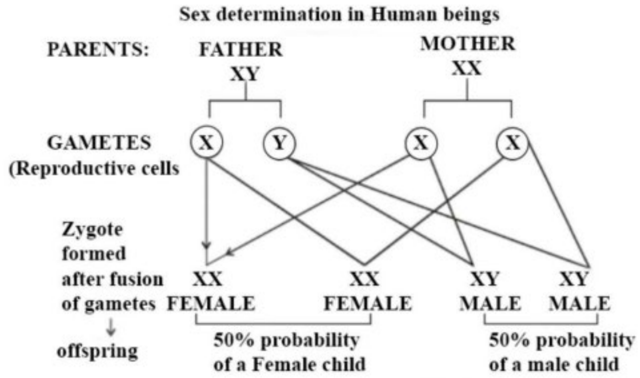
PRINCIPLE OF MENDELIAN INHERITANCE

I. **Law of Segregation:** The two alleles for each gene are placed in different gametes.

II. **Law of Independent Assortment:** The inheritance of one gene doesn't affect the inheritance of any other gene.

III. **Lzaw of Dominance**: When two different alleles are present, only one is dominant and will be expressed.

Sex Determination in human



Color blindness

Fundamentals of Genetics

- **Law of Independent Assortment:** Alleles for different traits are distributed to gametes independent of one another, provided the genes are on different chromosomes. Molecular Genetics & Modern Applications
- **Central Dogma:** The flow of genetic information in a cell typically goes from DNA to RNA to protein.
- The processes are replication (DNA to DNA), transcription (DNA to RNA), and translation (RNA to Protein).
- **Mutation:** A change in the nucleotide sequence of DNA.
- Mutations are a primary source of genetic variation and can lead to genetic disorders.
- **Genetic Disorders:** Diseases or problems caused by gene mutations or chromosomal abnormalities (e.g., sickle cell disease is an autosomal recessive disorder).
- **Biotechnology & Genetic Engineering:** Techniques involving the manipulation of DNA to alter genetic makeup, used in medicine and agriculture.
- **DNA Fingerprinting:** A method that uses the uniqueness of an individual's DNA to

establish identity, commonly used in forensic science and paternity testing.

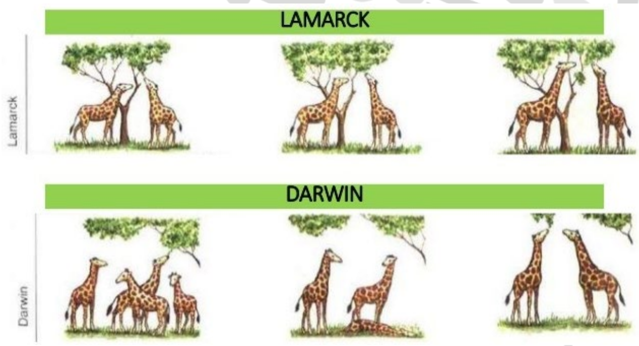
Evolution

Evolution is the gradual change in the heritable characteristics of biological populations over successive generations, driven by mechanisms such as natural selection and genetic variation.

Major Theories of Evolution:

- **Lamarck's Theory (Theory of Inheritance of Acquired Characters):** This early theory suggested that traits acquired by an organism during its lifetime (e.g., a giraffe's long neck from stretching) could be passed on to its offspring.
- This theory is largely discredited today.

Darwin's Theory (Theory of Natural Selection): Proposed by Charles Darwin and Alfred Russel Wallace, this is the most widely accepted theory.



Its core principles include:

- **Overproduction:** Organisms produce more offspring than can survive due to limited resources.
- **Variation:** Individuals within a species show natural, heritable variations in traits.
- **Struggle for Existence:** Competition for resources, predators, and mates creates a struggle for survival.
- **Survival of the Fittest (Natural Selection):** Individuals with advantageous traits for their specific environment are more likely to survive and reproduce, passing those traits to the next generation.
- **Adaptation:** Over time, these favourable traits become more common in the population, leading to adaptation to the environment.
- **Modern Synthetic Theory (Neo-Darwinism):** This integrates Darwin's natural selection with modern genetics.
- It emphasizes that genetic variation (due to mutation, genetic drift, and gene flow) provides the raw material upon which natural selection acts.

Evidence for Evolution

- **Fossils (Paleontology):** Fossils are the preserved remains of past life-forms found in rock layers.
- Studying the sequence of fossils in different geological strata shows how life has changed over time and documents intermediate forms (e.g., the evolution of the horse or human ancestors).

Anatomy (Homologous and Analogous Structures):

- Homologous Structures (e.g., the forelimbs of humans, bats, and whales) have a similar underlying structure but different functions, suggesting a common ancestry.
- Analogous Structures (e.g., the wings of a bird and an insect) have similar functions but different origins, showing convergent evolution

Evidence of Evolution

Evidences from comparative morphology and anatomy -

- Similarities and differences are found among organisms of today and those that existed years ago. Such similarities can be interpreted to understand whether common ancestors were shared or not.
- These similarities are of two types-
  (A) Homology (B) Analogy

(A) Homology -

Organs which have common origin, embryonic development and same fundamental structure but perform similar or different functions are called as Homologous organs and this phenomenon is called Homology.

Examples of homologous organs:

Forelimbs of mammals - Whales, bats, Cheetah and human (all mammals) share similarities in the pattern of bones of forelimbs though these forelimbs perform different functions. In these animals, forelimbs have similar anatomical structure - all of them have humerus, radius, ulna, carpals, metacarpals and phalanges in their forelimbs.

(a) Homologous structures (b) Analogous structures

Homologous and analogous structures are often confused and understanding them is of great importance in comprehending the similarities and differences between various organisms.

(B) Analogy -

The organs which have different origin and fundamental structures but perform similar functions are called Analogous organs and this phenomenon is called as analogy.

Examples of analogous organs:

(i) Wings of butterfly and birds - They are not anatomically similar structures though they perform similar functions i.e. used for flying.
(ii) Eye of the octopus and of mammals
(iii) Flippers of Penguins and Dolphins
(iv) Sweet potato (root modification) and potato (stem modification)
(v) Sting of bee and scorpion
(vi) Chloragogen cells of earthworm and liver of vertebrates.
(ii) Thorn of Bougainvillea and tendril of Cucurbita both are modification of axillary bud.
(iii) All vertebrate hearts
(iv) All vertebrate brain
(v) Mouth parts of insects -

Cockroach	**Honey bee**	**Mosquito**
(Biting & chewing)	(Chewing & lapping)	(Piercing & Sucking)

In each of these insects mouth parts comprise labrum, mandible maxilla etc.

(vi) Testes in male and ovaries in female
(vii) Potato and Ginger - both are modified shoot
(viii) Radish and Carrot - both are modified roots
(ix) Molecular homology - Homology found at molecular level. For example the plasma proteins found in the blood of man.

Evidence from Vestigial Organs

- The organs which are present in reduced form and do not perform any function in the body but are functional in related animals are called vestigial organs.
- They are remnants of organs which were complete and functional in their ancestors.

e.g. Nictitating membrane

- Muscles of pinna (auricular muscles)
- Vermiform appendix (Caecum)
- Coccyx
- Canine teeth
- Third molars (wisdom teeth)
- Body hair
- Nipples in males
- Segmented muscles of abdomen

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